

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: Applied Business Analytics Practical
Practical 5 . Probability and Probability Distributions:

AIM : Probability and Probability Distributions:

```
import numpy as np
import matplotlib.pyplot as plt

print("Name: Adarsh Shukla")
print("Roll No: T117")

rolls = 10000
dice = np.random.randint(1, 7, rolls)

print("Dice Rolling Experiment")
print("Total Rolls:", rolls)

for i in range(1, 7):
    count = np.sum(dice == i)
    probability = count / rolls
    print("Outcome", i, ": Count =", count, ", Probability =", probability)

plt.hist(dice, bins=np.arange(0.5, 7.5, 1), rwidth=0.8)
plt.title("Dice Roll Probability")
plt.xlabel("Dice Outcome")
plt.ylabel("Frequency")
plt.xticks(range(1, 7))
plt.show()

n = 10000

normal_data = np.random.normal(loc=50, scale=10, size=n)
uniform_data = np.random.uniform(low=0, high=100, size=n)
exponential_data = np.random.exponential(scale=10, size=n)

print("\nNormal Distribution")
print("Mean:", np.mean(normal_data))
print("Standard Deviation:", np.std(normal_data))

print("\nUniform Distribution")
print("Mean:", np.mean(uniform_data))
print("Standard Deviation:", np.std(uniform_data))

print("\nExponential Distribution")

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```

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```
print("Mean:", np.mean(exponential_data))  
print("Standard Deviation:", np.std(exponential_data))
```

```
plt.hist(normal_data, bins=30, edgecolor="black")  
plt.title("Normal Distribution")  
plt.xlabel("Value")  
plt.ylabel("Frequency")  
plt.show()
```

```
plt.hist(uniform_data, bins=30, edgecolor="black")  
plt.title("Uniform Distribution")  
plt.xlabel("Value")  
plt.ylabel("Frequency")  
plt.show()
```

```
plt.hist(exponential_data, bins=30, edgecolor="black")  
plt.title("Exponential Distribution")  
plt.xlabel("Value")  
plt.ylabel("Frequency")  
plt.show()
```

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```
a.py - C:/Users/IT-52/AppData/Local/Programs/Python/Python314/a.py (3.14.6)
File Edit Format Run Options Window Help

import numpy as np
import matplotlib.pyplot as plt

print("Name: Adarsh Shukla")
print("Roll No: T117")

rolls = 10000
dice = np.random.randint(1, 7, rolls)

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    count = np.sum(dice == i)
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plt.xticks(range(1, 7))
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print("Standard Deviation:", np.std(normal_data))

print("\nUniform Distribution")
print("Mean:", np.mean(uniform_data))
print("Standard Deviation:", np.std(uniform_data))

print("\nExponential Distribution")

plt.xticks(range(1, 7))
plt.show()

n = 10000

normal_data = np.random.normal(loc=50, scale=10, size=n)
uniform_data = np.random.uniform(low=0, high=100, size=n)
exponential_data = np.random.exponential(scale=10, size=n)

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print("Mean:", np.mean(normal_data))
print("Standard Deviation:", np.std(normal_data))

print("\nUniform Distribution")
print("Mean:", np.mean(uniform_data))
print("Standard Deviation:", np.std(uniform_data))

print("\nExponential Distribution")
print("Mean:", np.mean(exponential_data))
print("Standard Deviation:", np.std(exponential_data))

plt.hist(normal_data, bins=30, edgecolor="black")
plt.title("Normal Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()

plt.hist(uniform_data, bins=30, edgecolor="black")
plt.title("Uniform Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()

plt.hist(exponential_data, bins=30, edgecolor="black")
plt.title("Exponential Distribution")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
```

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Output :

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help

import matplotlib.pyplot as plt
ModuleNotFoundError: No module named 'matplotlib'

>>> ===== RESTART: C:/Users/IT-52/AppData/Local/Programs/Python/Python314/a.py =====
Name: Adarsh Shukla
Roll No: T117
Dice Rolling Experiment
Total Rolls: 10000
Outcome 1 : Count = 1750 , Probability = 0.175
Outcome 2 : Count = 1610 , Probability = 0.161
Outcome 3 : Count = 1667 , Probability = 0.1667
Outcome 4 : Count = 1635 , Probability = 0.1635
Outcome 5 : Count = 1654 , Probability = 0.1654
Outcome 6 : Count = 1684 , Probability = 0.1684

===== RESTART: C:/Users/IT-52/AppData/Local/Programs/Python/Python314/a.py =====
Name: Adarsh Shukla
Roll No: T117
Dice Rolling Experiment
Total Rolls: 10000
Outcome 1 : Count = 1639 , Probability = 0.1639
Outcome 2 : Count = 1643 , Probability = 0.1643
Outcome 3 : Count = 1639 , Probability = 0.1639
Outcome 4 : Count = 1734 , Probability = 0.1734
Outcome 5 : Count = 1660 , Probability = 0.166
Outcome 6 : Count = 1685 , Probability = 0.1685

Normal Distribution
Mean: 49.948173273457144
Standard Deviation: 9.989589392766584

Uniform Distribution
Mean: 49.66862609729161
Standard Deviation: 28.779831879928345

Exponential Distribution
Mean: 9.940755224119197
Standard Deviation: 10.078582169736288

>>>
```

```
===== RESTART: C:/Users/IT-52/AppData/Local/Programs/Python/Python314/a.py =====
Name: Adarsh Shukla
Roll No: T117
Dice Rolling Experiment
Total Rolls: 10000
Outcome 1 : Count = 1639 , Probability = 0.1639
Outcome 2 : Count = 1643 , Probability = 0.1643
Outcome 3 : Count = 1639 , Probability = 0.1639
Outcome 4 : Count = 1734 , Probability = 0.1734
Outcome 5 : Count = 1660 , Probability = 0.166
Outcome 6 : Count = 1685 , Probability = 0.1685

Normal Distribution
Mean: 49.948173273457144
Standard Deviation: 9.989589392766584

Uniform Distribution
Mean: 49.66862609729161
Standard Deviation: 28.779831879928345

Exponential Distribution
Mean: 9.940755224119197
Standard Deviation: 10.078582169736288

>>>
```

Graph :

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Figure 1

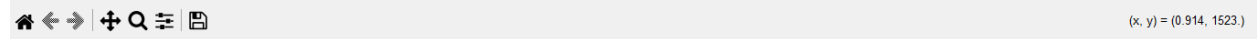
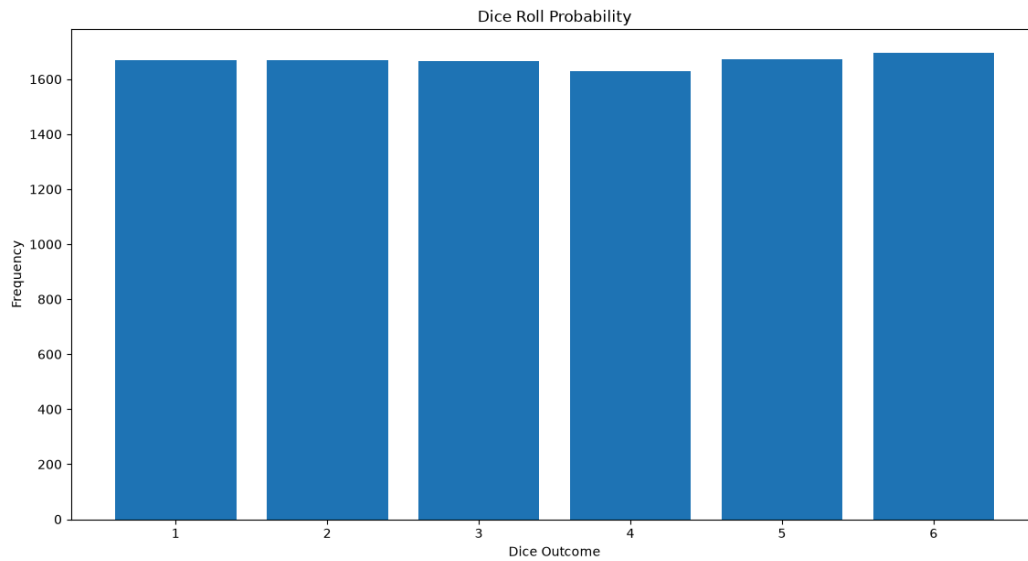
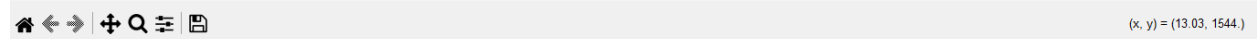
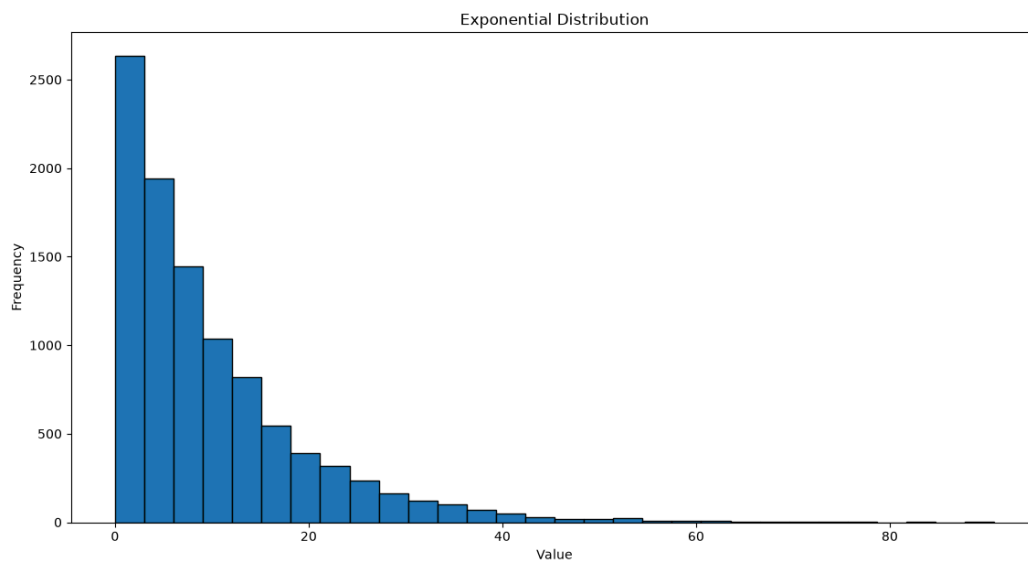


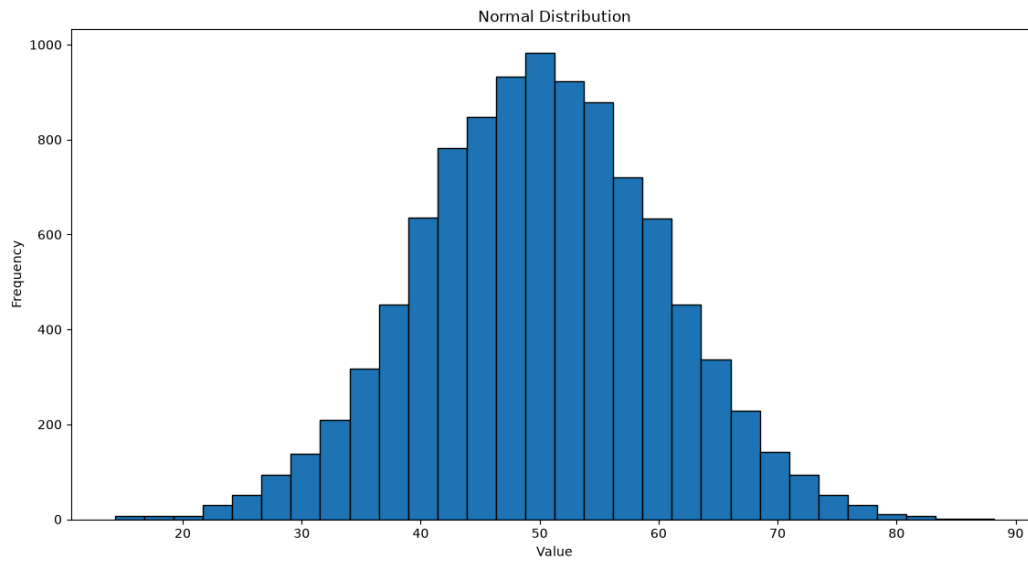
Figure 1



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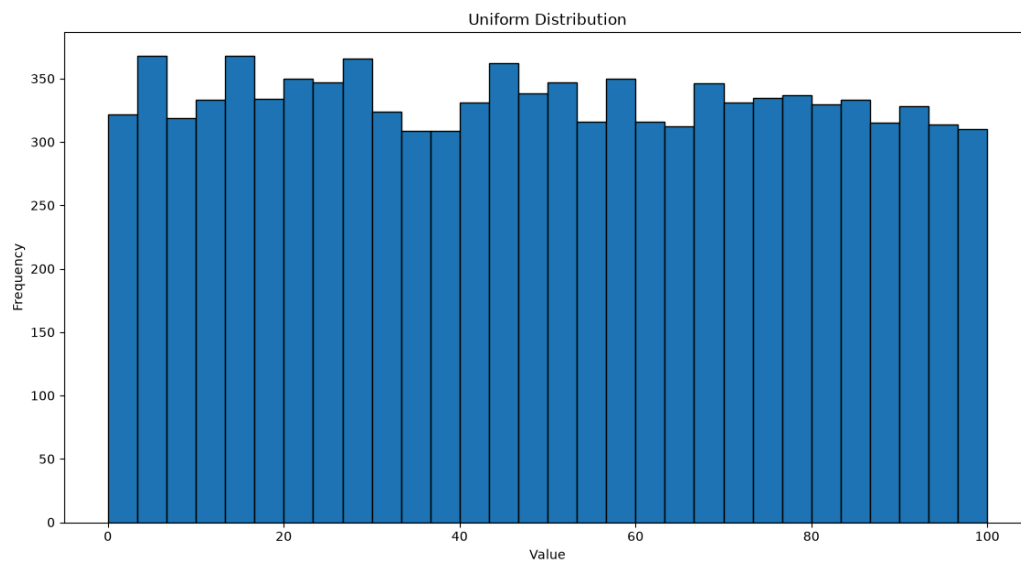
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Figure 1



(x, y) = (53.86, 815.)

Figure 1



(x, y) = (21.9, 212.0)

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